Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/31/2016
PI Name:	Eisch, Amelia Ph.D.		
Project Title:	HZE Particle Exposure-Induced Improvement of Pattern Separation in Mature Mice: Alterations in Mission-Relevant Behaviors and Neural Circuitry		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:	NOTE: Previously at University of Texas Southwestern Medical Center at Dallas, 2000-2016		
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-RADIATION
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No. of PhD Candidates:	5	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/17/2017 per S. Monk/HRP, due to PI move to Childrens' Hospital of Philadelphia; original end date was 1/29/2019 (Ed., 8/15/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	An unavoidable consequence of deep space missions is exposure to galactic cosmic radiation (GCR), which includes high (H) atomic number (Z) and energy (E) particles particles like Fe, Si, and O. Estimating radiation risks to the central nervous system (CNS) by HZE particles encountered during space missions is a high research priority. Past research has shown that rodents exposed to HZE particles have cognitive and performance deficits in numerous behavioral tasks, including those that rely on the hippocampus, a brain region involved in learning and memory. Notably, we have found that mature mice (of equivalent age to astronauts) exposed to either Si or Fe HZE particles actually show improved performance on a very difficult hippocampal task to assess the ability to discrimination two contexts that differ in discrete ways. Here we propose three aims to understand this improved context "pattern separation" after HZE particle exposure. In Aim 1, we hypothesize that HZE particle exposure-induced improved pattern separation is linked to improved performance on related learning and memory tasks, as well as executive function tasks, in the short-term, but to decreased performance in the long-term. In Aim 2, we hypothesize that HZE particle exposure-induced improved pattern separation is linked in the short-term to diminished stress-induced emergence of anxiety and depression-like behaviors, but to greater emergence in the long-term. In Aim 3, we hypothesize that HZE particle exposure-induced improved pattern separation is associated with disrupted hippocampal-cortical neural networks. All aims will rely on both classic and cutting-edge techniques. In sum, these aims will address whether the HZE particle exposure-induced improvement in pattern separation is beneficial or detrimental to mission success (Aims 1, 2), will indicate the integrity of neural circuitry contributing to mission-relevant behaviors (Aim 3), and will define both the short- and long-term health of neural networks needed to complete deep space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An unavoidable aspect of manned space flights is exposure to galactic cosmic radiation (GCR), which is made up primarily of protons (87%), followed by helium (11%) and then high atomic number (Z) and high-energy (HZE) particles (2%) like iron, silicon, and oxygen. The energy of HZE particles can be very high (>1000 MeV/u), sufficient in many cases to penetrate the spacecraft hull and interior materials, and they have a complex track structure and high linear energy transfer (LET). Thus, while the greatest physical radiation dose of GCR comes from high-energy protons, the greatest biological radiation dose of GCR comes from HZE particles. With long-duration and exploratory space missions in the near future, we need to understand how GCR influences human health and behavior. Estimation of radiation risks to the central nervous system (CNS) is a high research priority according to both a National Academy of Science report and NASA's Radiation Health Bioastronautics Roadmap. In rodents, HZE particles induce cognitive domain deficits, including decreased hippocampal learning and memory and cortically based executive function. However, it remains unclear whether the age at irradiation (IRR) influences the outcome of behavioral tests. It is also unclear whether these cognitive decrements extend to other cognitive tests, or to other behavioral domains, such as mood and stress response. Recently, we found that mice exposed at maturity ("astronaut-aged") to 28Si or 56Fe HZE particle IRR perform better than control mice on a hippocampal-based pattern separation task (context discrimination fear conditioning, CDFC). We want to understand if this behavioral improvement in pattern separation is reflective of other changes in behavior, and whether these changes will be beneficial or detrimental to mission success. We propose a behavioral domain- and brain network-based analysis to understand the HZE particle-induced behavioral improvement shown in our pilot data. These data will have relevance for understanding the risks facing crew members in deep space missions, particularly in regards to the age of crew member at the time of the mission.
	Using mice that are irradiated at "astronaut-age" and multiple time points post-IRR, we will determine: Aim 1: The impact of HZE particles on other tests in the cognitive domain. Using the "flexible battery" of tests offered by the touchscreen operant platform which we have received, we are in the process of identifying a behavioral framework for the HZE particle-induced improvement in pattern separation. We have hired experienced staff and have a series of small, discrete experiments are in progress in which we will evaluate the effectiveness of our selected battery. Then we will irradiate our first group of HZE mice in Summer 2016, and our first touchscreen data after HZE irradiation will be available in Fall 2016. We hypothesize HZE particles will improve performance reliant on hippocampal-cortical integrity, but not on striatal-cortical integrity. Aim 2: The impact of HZE particles on behaviors in the affective/social domain. Pattern separation is important not just to contextual discrimination, but also to the resistance to mood and anxiety disorders. Based on this, and with previous experience with an ethologically relevant stressor, we will test HZE particle-induced influence in the affective/social domain. We have secured experienced personnel and planned our future irradiation runs. We have laid out our plans for progress even while the main lab transfers academic institutions (Sept 2016), and are coordinating with lab personnel to minimize down time. When we perform these studies in 2017, we hypothesize that improved pattern separation is linked to resilience to social stressors. Aim 3: Indices of neural network perturbation underlying HZE particle exposure-induced improved pattern separation. While improved pattern separation has been causally linked to increased hippocampal neurogenesis, our data show HZE particle-induced improvement in pattern separation is actually marked by decreased hippocampal neurogenesis. Initial results also show that interneuron number is not altered in irradiated animals vs. sham animals. We are currently using cutting-edge techniques (RNAseq) to test the hypothesis that HZE particles influence the integrity of hippocampal-cortical neural networks. Publications supported by this grant (2) 1. Latchney SE*, Jiang Y*, Petrik D, Eisch AJ^, Hsieh J^. Inducible knockout of Mef2A/C/D from nestin-expressing stem/progentior cells and their progeny unexpectedly uncouples neurogenesis and dendritogenesis in vivo. The Journal of the Federation of American Societies for Experimental Biology (The FASEB Journal). 2015 Aug 18. pii: fj.15-275651 13pp. PMID 26286136, PMC in progress. *equal contribution, ^co-corresponding authors 2. Petrik D, Latchney SE*, Masiulis I*, Yun S*, Zhang Z*, Wu JI^, Eisch AJ^. Chromatin remodeling factor Brg1 supports the early maintenance and late responsiveness of nestin-lineage adult neural stem and progenitor cells. Stem Cells. 2015 Sep 29. doi: 10.1002/stem.2215. 11pp. PMID 26418130, PMC in progress. *equal contribution, listed alphabetically, ^co-corresponding authors Manuscripts supported by this grant in revision, preparation, or progress (10)

Task Progress:	1. Rivera PD, Reynolds RP, Just AL, Birnbaum SG, Eisch AJ. Image-guided cranial irradiation-induced ablation of dentate gyrus neurogenesis diminishes extinction of young – but not old – morphine reward memories. In revision for resubmission to Hippocampus. 2. Bulin SE, Richardson DR, Mendoza ML, Zhang Z, Song KH, Solberg TD, Eisch AJ. Ablation of dentate gyrus neurogenesis via cranial irradiation enhances morphine self-administration and morphine-induced locomotor sensitization. In revision for resubmission to Biological Psychiatry. 3. Walker AK*, DeCarolis NA*, Ahn HF, Lagace DC, Ables JL, Redfield RL, Cole MG, Chen BPC, Eisch AJ. Running after irradiation ameliorates the proliferation deficit but not the nestin-expressing Type-1 cell deficit in the adult hippocampal subgranular zone. In revision for resubmission to Hippocampus. *equal contribution 4. Norbury JW, Schimmerling W, Slaba TC, Azzam E, Badavi FF, Baiocco G, Benton E, Bindi V, Blakely EA, Blattnig SR, Boothman DA, Borak1 TB, Britten RA, Curtis S, Dingfelder M, Durante M, Dynan W, Eisch AJ, Elgart SR, Goodhead DT, Guida PM, Heilbronn LH, Hellweg CE, Huff JL, Kronenberg A, La Tessa C, Lowenstein D, Miller J, Morita T, Narici L, Nelson GA, Norman RB, Ohnishi T, Ottolenghi A, Patel ZS, Reitz G, Adam Rusek A, Schreurs A-S, Scott-Carnell LA, Semones E, Shay JW, Shurshakov VA, Sihver L, Simonsen LC, Story M, Turker MS, Uchihori Y, Williams J, Zeitlin CJ. Galactic cosmic ray simulation at the NASA Space Radiation Laboratory. Under review, Life Sciences in Space Research 5. Walker AK*, Whoolery CW*, Richardson DR, Reynolds RP, Rivera PD, Shih H-Y, Redfield RL, Lucero MJ, Beddow DH, Mukherjee S, LeBlanc JA, Cole MG, Amaral WZ, Malhotra S, DeCarolis NA, Chen BPC, Eisch AJ. Whole body exposure to high LET 28Si particles decreases hippocampal progenitor proliferation and neurogenesis in the short-term, and new neuron survival in the long-term. In preparation for submission to Radiation Research. *equal contribution, listed alphabetically. 6. Lucero MJ,* Walker AK,* Whoolery CW,* Redfield RL, Ito N, Richardson DR, Reynolds RP, Palchik G, Mukherjee S, Shih H-Y, Rivera PD, Birnbaum SG, Chen BPC, Eisch AJ. Space radiation unexpectedly enhances pattern separation yet diminishes dentate gyrus neurogenesis in a dose- and age-dependent manner. In preparation. *equal contribution, listed alphabetically 7. Reynolds RP, Yun S, Eisch AJ. From synapse to synchrony: distinct dentate gyrus interneurons orchestrate and refine local dentate gyrus circuitry. In preparation. 8. Yun S, Lucero MJ, Ito N, Rivera PD, Birnbaum SG, Chen BPC, Eisch AJ. Molecular basis for space radiation-induced deficits hippocampal function in aged animals. In progress 9. Cole MG, Shih H-Y, LeBlanc JA, Rivera PD, Chen BPC, Eisch AJ. Short and long term effects of 16O radiation on adult hippocampal neurogenesis and indices of genomic instability. In progress. 10. Cole MG, Shih H-Y, LeBlanc JA, Chen BPC, Eisch AJ. A model of acute space radiation exposure (12C) and adult hippocampal neurogenesis. In progress Presentations/published abstracts (3) 1. Lucero MJ, Redfield RL, Ito N, Reynolds RP, Richardson DR, Palchik G, Mukherjee S, Shih HS, Rivera PD, Birnbaum SG, Chen BPC, Eisch AJ. Neural Mechanisms Contributing to HZE Radiation-Induced Improvement in Pattern Separation in Adult Mice. NASA 2015 Human Research Program Investigators' Workshop, Galveston TX, February 2015. 2. Walker AK*, Whoolery CW*, Richardson DR, Reynolds RP, Rivera PD, Shih H-Y, Redfield RL, Lucero MJ, Beddow DH, Mukherjee S, Chen BPC, Eisch AJ. Galactic cosmic radiation (28Si) reduces dentate gyrus neurogenesis in the long -term in a dose -dependent manner. The Annual Meeting of the Society for Neuroscience, Chicago, IL October 2015. *equal contribution, listed alphabetically. 3. Eisch AJ, Lucero MJ, Redfield RL, Ito N, Richardson DR, Reynolds RP, Palchik G, Mukherjee S, Walker AK, Whoolery CW, Shih H-Y, Rivera PD, Birnbaum SG, Chen BPC. What happens to mouse hippocampal--dependent behavior and neurogenesis on the way to Mars? One small step for mousekind. The Annual Meeting of the Society for Neuroscience, Chicago, IL October 2015.
Bibliography Type:	Description: (Last Updated: 11/17/2025)
Articles in Peer-reviewed Journals	Latchney SE*, Jiang Y*, Petrik D, Eisch AJ^, Hsieh J^. (*equal contribution, ^co-corresponding authors) "Inducible knockout of Mef2a, -c, and -d from nestin-expressing stem/progenitor cells and their progeny unexpectedly uncouples neurogenesis and dendritogenesis in vivo." FASEB J. 2015 Dec;29(12):5059-71. Epub 2015 Aug 18. http://dx.doi.org/10.1096/fj.15-275651 ; PubMed PMID: 26286136; PubMed Central PMCID: PMC4653059 , Dec-2015
Articles in Peer-reviewed Journals	Petrik D, Latchney SE*, Masiulis I*, Yun S*, Zhang Z*, Wu JI^, Eisch AJ^. (*equal contribution, listed alphabetically, ^co-corresponding authors) "Chromatin remodeling factor Brg1 supports the early maintenance and late responsiveness of nestin-lineage adult neural stem and progenitor cells." Stem Cells. 2015 Dec;33(12):3655-65. Epub 2015 Oct 9. http://dx.doi.org/10.1002/stem.2215 ; PubMed PMID: 26418130; PubMed Central PMCID: PMC4713255 , Dec-2015